

ARYA ZADEBRAHIMI

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PERSONAL WEBPAGE

EDUCATION

Aug 2024

Master of Science in Bioinformatics

Sharif University of Technology

- Cumulative GPA of 3.9 (18.2 / 20)
- Awarded a scholarship after ranking 36th out of 13,000 in the national university entrance exam.
- Relevant coursework: Synthetic Biology, Machine Learning for Bioinformatics, Cell and Molecular Biology, Structural Bioinformatics, Computational Genomics

May 2021

Bachelor of Science in Software Engineering

Azad University

- Cumulative GPA of 3 (15.58 / 20)
- Selected coursework: Probability and Statistics, Artificial Intelligence, Data Mining, Algorithms Design, Computer-aided design of digital systems

RESEARCH INTEREST

- Microbial Communities
- Metabolic Engineering
- Graph Neural Networks

RESEARCH EXPERIENCE

Synthetic auxotrophic community design with maximum growth rate

- A constrained optimization model to find an auxotrophic community of any size.
- This model is based on previous work, SteadyCom, with additional constraints to ensure auxotrophic behavior.
- It ensures robustness through modular behavior; each community member produces a unique metabolite required for survival.
- Failed due to high implementation complexity of one constraint that required another optimization.

Designing a synthetic community to increase by-product yield

- A constrained optimization model to find the fittest community to maximum any desired by-product yield.
- This model is based on previous work, SteadyCom, with additional constraints to search through microorganisms to find the fittest ones.
- Failed due to the high computational cost for large metabolic networks.

Predicting metabolic reaction Flux using Graph neural networks

- A Novel approach to predict metabolic flux using deep neural networks that reduce computational cost dramatically.
- A large disconnected graph representing several small metabolic networks for each microorganism.
- Achieved a Root Mean Square Error of 4 between predicted flux and Flux Balance Analysis results.

PROJECTS

- Analysis of microarray gene expression data to identify genes involved in AML (Acute Myeloid Leukemia)
- Docking and optimizing the conformation of the Fruquintinib inhibitor molecule
- A transformer-based model to predict financial market price
- Single-cell RNA sequencing clustering and analysis

TEACHING ASSISTANT

Generative Model Design

- Assignment Design
- Test Question Design

Biomedical Image Processing

- Assignment Design
- Test Question Design

FURTHER EXPERIENCE

Aug 2022

Wet Lab Techniques and Molecular Biology

Iran University of Medical Sciences, Tehran

- Three-day internship
- Cell DNA extraction
- DNA detection using PCR
- Preparing medium and cell culture
- Cell counting using microscope

Oct 2017 – Dec 2018

Android Developer

Tehran

- Developed 5 separate applications
- Agile project management methodology

SKILLS

TOEFL: 104 (Reading 30, Listening 28, Speaking 23, Writing 23)

Programming: Python (pytorch, pytorch geometric, transformers, gymnasium, numpy, pandas, opencv), Julia, Java, R, Android, Verilog, TCL, C#, React